

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer — AI/ML Engineer — Remote

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Professional Summary

Motivated Lead Data Scientist with a proven track record of delivering over \$1.3 billion in business value through machine learning and end-to-end data science solutions for complex challenges. Expert in managing the software development lifecycle, emphasizing stakeholder engagement, KPI definition, team building, and scalable deployment. Ability to articulate complex data insights to diverse stakeholders, ensuring clear understanding and informed decision-making.

Education

Bachelor of Science in Mechanical Engineering – Arizona State University — Tempe, AZ — 06/2006

Hard Skills

A/B Testing Agile Methodology, AI/ML, Anomaly Detection, Architectural Design, Artificial Intelligence, AWS, CI/CD, Computer Science, Continuous Integration, DAG, Database Administrator, Data Gathering, DBA, Deep Learning, DevOps, Developer Tools, Distributed Computing, Docker, ECR, Embedding, ETL, Financial Forecasting, Git, GitHub Actions, GitHub Copilot, Linear Algebra, Large Language Models, LLMs, Mathematics, Multivariate Calculus, Pandas, Power BI, Problem Solving, PyTorch, Redshift, Research, Scikit-learn, S3, Sagemaker, Scripting Language, Scrum, SDLC, SnapLogic, Snowflake, SQL, Step Functions Streamlit, Tableau, TensorFlow, Transformers, Version Control

Soft Skills

Adaptability, Business Requirements, Coaching, Communication Skills, Creative, Growth Mindset, Leadership, Mentoring, Multicultural Collaboration, Strategic Thinking, Thought Leadership, Willing to Learn

Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ — 09/2022 to Present

- Architected and developed end-to-end data science initiatives, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award.
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication.
- Wrangled structured, semi-structured, and unstructured data to assemble a training dataset from multiple sources, considering thousands of parameters derived from dealer and product information. Used an LLM to find similarities in job descriptions and feature engineering to retain important parameters, enhancing model accuracy.
- Conducted exploratory data analysis on the training dataset, eliminating outliers using sophisticated analytics and anomaly detection techniques. Utilized Python and Autogluon to train the AI model, leveraging machine learning frameworks such as neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, improving model performance.
- Engineered and deployed robust MLOps pipelines using AWS and Snowflake, as a database administrator, and established best practices for cloud and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact.

Reliability Analytics Manager — Amazon.com — Tempe, AZ — 02/2022 to 09/2022

- Pioneered design engineering KPI for autonomous robots by analyzing petabyte-scale IoT big data. Authored an influential white paper, as reproducible research using R, on the framework for performing statistical modeling and secured adoption by senior leadership.
- Led team of business intelligence engineers and data analysts to deploy pipeline which communicated the status of robots at all global fulfillment centers using Shiny R.
- Designed and optimized regression and recommendation systems, achieving two-time champion status in Amazon Machine Learning University competitions, which improved predictive accuracy and recommendation quality in practical applications

Data Engineering Manager — Empire-Cat — Mesa, AZ — 01/2012 to 02/2022

- Created ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days.
- Developed and deployed an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting.
- Directed a team of up to 8 engineers and served as a client-facing expert for mining equipment, fostering a data-driven approach to solve equipment performance issues, which improved team efficiency and client satisfaction.
- Applied the DMAIC Six Sigma methodology with complex problem-solving using R and SQL to identify opportunities for improving operation, manufacturing, and maintenance of a large off-highway truck fleet, yielding \$1.2 billion in savings over six years and significantly enhancing customer experience.

Certifications

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **CSCMP:** Supply Chain Foundations Demand Planning Professional Certificate
- **Data Camp:** AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate
- **Empire-Cat:** Business Tools for Service Managers, Coaching for Peak Performance, Communicating for Leadership Success, Delegating with Purpose, Lean 6 Sigma Green Belt, Resolving Workplace Conflict, Setting Performance Expectations
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IUOE Local 150:** Diesel Mechanic, Bulldozers, Class B CDL, Cranes, Forklifts, Heavy Equipment Operator, MSHA, OSHA, Scrapers, Skid-steers, Telehandlers, Wheel Loaders Shielded Metal Arc Welding (SMAW)
- **Komatsu North America:** Certified Technical Communicator
- **LinkedIn Learning:** Data Engineering with dbt
- **Society of Automotive Engineers:** Cost and Finance for Engineers
- **Udemy:** Streamlit for Snowflake Masterclass